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Attention-PestNet: hierarchical scaled dot-product attention for insect pest detection

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Abstract

Timely and precise insect pest detection is critical in areas with high agricultural intensity and climates that favor continuous pest activity. Traditional pest identification methods, such as manual inspection or expert guided analysis, are labor intensive and time consuming. These approaches lack scalability and hinder timely intervention, particularly in resource-constrained settings. Furthermore, the high visual similarity between pest species and intra-species variability across developmental stages further challenge detection efforts in real-world agricultural conditions. To address these limitations, we propose Attention-PestNet, a novel one-stage object detection network designed for insect pest detection. Our method consist of two key attention-based modules to enhance feature extraction and improve detection performance. First, the Hierarchical Scaled Dot-Product Attention module leverages a multi-level attention mechanism to capture salient pest features at different scales. Second, the Multi-Scale Spatial Attention module refines spatial feature representations by incorporating horizontal and vertical attention pathways with multi-scale max-pooling operation to enhance contextual understanding. Extensive experiments were conducted on two public benchmarks, IP102 and R2000 datasets, which represent agricultural conditions in Asia. The results demonstrate that Attention-PestNet outperforms state-of-the-art models in both visualization outputs and quantitative metrics. Attention-PestNet shows strong potential as a scalable and cost-effective solution for intelligent pest monitoring in modern precision agriculture. Our code and data for this paper are made available at: <https://github.com/thinhdoanvu/HSDPA>.

Keywords Hierarchical scaled dot-product attention, Multi-scale spatial attention, Attention-based object detection, Insect pest detection

Introduction

Insect pests pose a persistent and serious threat to global agriculture by causing extensive crop losses and undermining food security. As of 2025, insect pests and plant diseases cause up to 40% of global crop losses annually, with estimated economic damages exceeding USD 220 billion [1]. These impacts are particularly severe in regions dominated by smallholder farming systems, where limited access to timely pest management exacerbates food insecurity risks. According to the Food and Agriculture Organization (FAO), smallholder farmers cultivate approximately 80% of the agricultural land in Asia, where over 200 million hectares of maize and rice

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are grown annually. Meanwhile, more than 90% of the world's rice is produced and consumed in this region. Asia is particularly vulnerable to pest outbreaks due to a combination of conducive climatic conditions, intensive cropping systems, and the prevalence of smallholder farming. The region's tropical and subtropical climate supports rapid pest reproduction and year-round survival. These factors foster favorable conditions for pest survival and proliferation, enabling year-round reproduction and infestation cycles.

Insect pests cause direct damage to crops by feeding on leaves, stems, roots, and fruits. Indirectly, they diminish crop yields, degrade product quality, and increase production costs [2]. Most pest control methods are based on manual inspection, pheromone traps [3], and plant and soil analysis [4]. While widely adopted, these conventional methods are often limited by high labor demands, delayed response times, and dependence on expert intervention [5]. The limitations of existing approaches underscore the critical need for innovative, scalable, and cost-effective pest monitoring solutions, particularly in vulnerable smallholder systems.

Traditional machine learning methods, notably Support Vector Machines (SVMs) [6] and random forests [7], heavily rely on handcrafted features for pest detection, which require substantial expertise and experience. Furthermore, pest images captured in real agricultural environments are typically exhibit complex backgrounds and lighting variations, which may affect the detection accuracy. In contrast, deep learning approaches, notably Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), offer significant improvements over traditional approaches by autonomously learning hierarchical feature representations from large-scale image data. This capability enables more accurate and robust identification of pests in complex and variable field conditions. As a result, deep learning-based frameworks have become increasingly prevalent in the development of automated pest monitoring solutions [8]. For example, Li et al. [9] employed two typical two-stage methods, Faster R-CNN and Mask R-CNN, trained on the IP102 benchmark [10] for detecting 10 categories of insect pests. He et al. [11] developed a two-layer detection architecture using Faster R-CNN to recognize *Nilaparvata lugens* Stal. Similarly, Deng et al. [12] proposed a federated learning framework combined with an improved Faster R-CNN to simultaneously detect multiple orchard pests and crop diseases. Despite their accuracy, these models significantly challenged by the small dense distribution and complex backgrounds of insect pests. In addition, the intensive computational cost limits these methods in real-time applications where prediction time is critical. To address real-time limitations, several one-stage pest detection methods have

been proposed using the IP102 dataset. For example, Wei et al. [13] evaluated AEC-YOLOv8n, which achieved mAP@0.5 of 67.1%. Fan et al. [14] applied a Squeeze-and-Excitation (SE) module to YOLOv5s, yielding mAP@0.5 of 35.5%. Xiang et al. [15] proposed the YOLO-Pest network with a ConvNext module inspired by the Vision Transformer, with mAP@0.5 of 57.1%. Yang et al. [16] proposed YOLO-RP, a lightweight and efficient rice pest detection method based on YOLO11n, which recorded 63.84% mAP@0.5. In another study, Zheng et al. [17] presented a lightweight rice pest detection approach built upon an improved YOLOv8 backbone, trained on a combined dataset derived from selected R2000 images, and reported 70.2% mAP@0.5. Although these one-stage methods improve efficiency, they still face challenges in detecting small pests under complex backgrounds and varying illumination, which are common in real-world agricultural scenarios [18]. Specifically, two major challenges persist: (i) high visual similarity among different pest species (Fig. 1a), which can mislead classification models; and (ii) substantial appearance variation within the same species across developmental stages, such as larvae, pupae, and adults (Fig. 1b), which hinders consistent identification.

Recent work has emphasized the importance of explicitly modeling the variations caused by different life stages. Han et al. [19] proposed a stage-aware multi-task framework that simultaneously learns pest species and their developmental stages to enhance robustness against intra-species variation. The method incorporates multi-stage co-supervision, which allows the network to capture both stage-specific features and the inherent similarities within a species while distinguishing between different species. Additionally, the framework employs a spatial attention mechanism to improve the model's ability to localize and extract discriminative regions, thereby reducing misclassification caused by inter-species similarity. Similarly, several studies have leveraged multi-scale representation learning to handle pests with diverse sizes and visual patterns. Wang et al. [20] improved U-Net architecture, termed DMSAUN-Net. The model integrates a dilated Inception module for multi-scale contextual feature extraction, a spatial attention mechanism to enhance edge focus and reduce noise, and ResNet-based skip connections to further lower computational cost while maintaining performance. Conversely, Linfeng et al. [21] designed a multi-branch information-sharing network with fine-grained attention to solve the problem of high inter-class similarity. Despite these advances, existing approaches still struggle when pests exhibit extreme morphological changes across developmental stages or when multiple species share highly similar appearance under real-world field conditions.

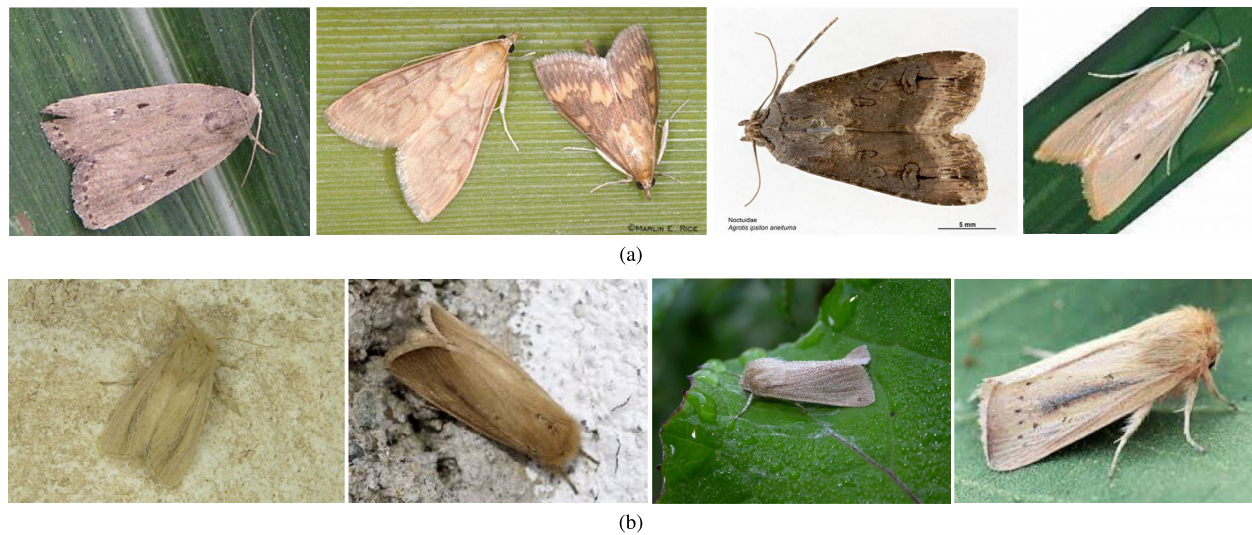


Fig. 1 Illustration of two challenges in pest detection. Top row: High visual similarity among yellow cutworm, corn borer, black cutworm, and Asiatic rice borer (arranged from left to right). Bottom row: Multiple developmental stages of the *Mythimna pallens*. Best viewed in color

In this paper, we address the aforementioned research gaps by proposing Attention-PestNet, a one-stage object detection framework specifically designed for pest detection. Our method introduces two attention-based modules to enhance feature extraction and improve detection performance: (i) a *Multi-Scale Spatial Attention* (MSPA) module that captures multi-scale spatial relationships to better distinguish small pests from background clutter; (ii) a *Tri-Directional Attention* (TDA) module that jointly models height-wise, width-wise, and channel-wise correlations to refine feature representations and support more precise detection of visually similar pests, while reducing the computational cost compared to conventional self-attention; and (iii) a *Hierarchical Scaled Dot-Product Attention* (HSDPA) module that captures long-range global dependencies to enrich semantic context across feature maps. This attention-based design enables Attention-PestNet to detect a wide variety of crop pests effectively and achieve state-of-the-art accuracy on the IP102, R2000, and Pest24 benchmark datasets.

In the MSPA module, spatial attention is combined with multi-scale pooling to effectively capture spatial relationships at multiple scales instead of the commonly used single-scale feature extraction. Specifically, global and max-pooling operations extract fine contextual features, while a convolutional layer refines the representations and reduces channel dimensions. In the HSDPA module, the TDA is applied multiple times to enhance feature representation. To improve efficiency and representational capacity over traditional sequent designs, we introduce a hierarchical structure that concatenates the outputs of each TDA block with the original input. A 1×1 convolution is subsequently applied to compress

the combined features, which results in improved performance and computational cost.

The main contributions of this paper can be highlighted as follows.

- We introduce Attention-PestNet, a novel one-stage object detector specifically designed for insect pest detection. Our method achieves a balance between computational efficiency and detection accuracy by integrating multi-scale spatial and hierarchical attention mechanisms, allowing the network to capture both global and local contextual information while maintaining practical inference speed for real-world agricultural applications.
- We propose the MSPA module, which combines spatial attention with multi-scale pooling to improve feature extraction. This design enables the network to effectively learn spatial relationships and capture multi-scale features at different scales, leading to more accurate recognition of complex patterns and objects of varying sizes.
- We propose the TDA module, which extends the Scaled Dot-Product attention to three directions—height, width, and channel. This structure captures long-range dependencies across both spatial and channel dimensions while maintaining computational efficiency.
- We propose the HSDPA module, which applies the TDA modules in a hierarchical manner to extract diverse feature representations. The outputs of the Scaled Dot-Product mechanism are concatenated with the original input to enrich the feature space. This effectively improves the network's ability to understand the relationships between different

features and focus on salient spatial regions within the input.

- We conducted extensive experiments on three public benchmark datasets: IP102, R2000, and Pest24. The state-of-the-art performance achieved by our Attention-PestNet demonstrates its effectiveness and robustness across diverse conditions.

The remainder of this paper is organized as follows. Section 2 provides a review of related work on pest detection methods. Section 3 describes the proposed attention-based detection method. Section 4 presents experimental evaluations of the major methods on the IP102, R2000, and Pest24 datasets. Section 5 provides a comprehensive discussion on the performance, limitations, and potential improvements of the proposed architecture. Finally, section 6 provides concluding remarks.

Related work

This section provides a review of two categories of methods relevant to pest detection: generic object detection techniques in computer vision (Section 2.1) and specialized approaches for insect pest detection (Section 2.2).

Generic object detection methods

Object detection is a key computer vision task involving the identification and localization of objects within an image. With the advancement of deep learning, object detection frameworks are primarily categorized into two types: two-stage detectors and one-stage detectors. Two-stage object detectors, notably the Region-based Convolutional Neural Network (R-CNN) family [40–42], operate in a sequential pipeline comprising two main stages. First, region proposals are generated using methods such as Selective Search [40] or a Region Proposal Network (RPN) [41]. These candidate regions are then passed through a deep neural network to extract features. Second, the extracted features are processed by two parallel branches: one performs classification to identify object categories, while the other refines bounding box coordinates to improve localization accuracy.

One-stage object detectors, notably the SSD [43] and YOLO series, directly predict class possibilities and bounding box offsets for each spatial location of the full image without region proposals generation. To handle multiple overlapping predictions and enhance detection accuracy, post-processing techniques such as Non-Maximum Suppression (NMS) [44] are employed to eliminate redundant bounding boxes by retaining those with the highest confidence scores. The One-to-One Head mechanism in YOLOv10 [45] replaces NMS by enforcing a one-to-one assignment between predictions and ground truth, to reduce complexity and improve training efficiency.

The evolution of YOLO versions demonstrates continuous improvements in object detection performance. From YOLOv1 to YOLOv3 [46–48], the focus was on enhancing accuracy through techniques such as multi-scale training and residual connections. YOLOv4 [49] and YOLOv5 [50] introduced advanced feature fusion techniques such as PANet and SPP, while also improving computational efficiency. YOLOv6 [51] marked a shift towards anchor-free models, while YOLOv7 [52] returned to anchor-based detection with additional network improvements. The subsequent versions, YOLOv8 [53], YOLOv9 [54], YOLOv10 [45], YOLOv11 [55], and YOLOv12 [56], further refined the architecture with innovations like anchor-free heads and the incorporation of CNN-Transformer hybrid architectures, leading to improved flexibility, efficiency, and performance in a wide range of object detection tasks. In [43], Li et al. introduced SSD, an object detection framework that employs a CNN as its backbone for feature extraction. SSD utilizes multiple layers from the base network to generate multi-scale feature maps that allow object detection across different object sizes. At each feature map level, SSD assigns anchor boxes with different aspect ratios and sizes while simultaneously predicting object class scores and bounding box offsets.

Recent studies have applied the transformer architecture to object detection. The DETection TRansformer (DETR) [57], was the first end-to-end object detection framework to adopt a transformer encoder-decoder architecture. Instead of relying on predefined anchor boxes as in YOLO and Faster R-CNN, DETR learns object locations using a transformer-based global loss. The architecture consists of a CNN backbone for feature extraction, followed by a transformer encoder that captures global relationships between objects. The transformer decoder then uses the learned object queries to generate predictions, while a feedforward detection head assigns class labels and refines bounding boxes.

Pest detection methods

Agricultural pest detection is a challenging fine-grained recognition task due to the large intra-species variation (e.g., different developmental stages, varying shapes and colors, environmental conditions) and strong inter-species similarity among morphologically related pests. Numerous studies have attempted to address these challenges by designing models capable of capturing subtle discriminative features while remaining robust to appearance variability.

Traditional image processing techniques have been widely used in early pest detection models. In [58], Xia et al. proposed an automatic pest identification method for whiteflies, aphids, and thrips. The method proceeds in two stages: first, insects are segmented from sticky

trap backgrounds using the watershed algorithm; second, color features are extracted via Mahalanobis distance in the YCrCb color space.

Two-stage object detection methods remain a widely used strategy in recent pest detection models. Due to the small size of certain insects, traditional image processing techniques may not produce accurate results. To address this issue, Nanni et al. [38] employed CNNs with multiple backbone architectures, including ResNet50 [59], GoogleNet [60], ShuffleNet [61], MobileNetV2 [62], and DenseNet201 [63]. These networks were optimized using Adam variants and two novel DGrad-based optimizers: Exponential DGrad and Exponential Learning Rate, which adapt learning rates using gradient variation patterns. Similarly, Ali et al. [64] proposed a region-based deep learning model for large-scale multiclass pest detection. The model consists of three main components: (i) a Channel-Spatial Attention module integrated into the CNN backbone to enhance feature extraction, (ii) a RPN for generating pest candidate regions, and (iii) a Position-Sensitive Score Map module for classification and bounding box regression. Additionally, Contextual RoI is employed to incorporate surrounding context and improve detection accuracy. Xiao et al. [65] introduced an agricultural pest detection algorithm based on an improved Faster R-CNN framework. Specifically, DenseNet169 and a scale-transfer module are utilized to extract multi-scale feature maps, while the RPN generates precise bounding boxes, followed by a fully connected layer for final pest classification.

One-stage object detection methods have proven effective in pest detection. In [66], Qiang et al. improved the SSD framework with a double backbone network that combines two standard feature extractors, with VGG16 [67] serving as the primary backbone and ResNet50 [59] as the secondary. To improve feature representation, the lower layer features of ResNet50 are progressively fused with the corresponding layers of VGG16. This approach enhances small-object detection through multi-scale feature fusion and anchor-based predictions. In [68], Chu et al. proposed a multi-scale granary pest recognition model based on the YOLOv5 object detection algorithm. This approach integrates three key modules: Efficient Channel Attention mechanism (ECA), Bidirectional Feature Pyramid Network (BiFPN), and Distance Intersection over Union Non-Maximum Suppression (DIOU-NMS). Specifically, the ECA mechanism is incorporated into the feature extraction module to enhance the model's ability to capture small target information; the BiFPN is adopted in the feature fusion module to improve the flow of information between higher and lower layers; and DIOU-NMS is employed in the prediction module to replace the traditional NMS, effectively addressing the issue of overlapping bounding boxes caused by densely

packed targets. Yang et al. [32] developed a YOLOv7-based detection model tailored for pest monitoring in tea plantations. The model integrates deformable convolutions and a dynamic attention mechanism to enhance feature extraction. The authors also introduced a novel implicit decoupling head and replaced the standard NMS algorithm with a softened version to improve prediction accuracy.

Recently, transformer-based architectures have been explored for pest detection. In [69], Li & Lee introduced a Transformer-based approach to detect and prevent citrus greening disease. It utilizes the Cascade R-CNN Swin model, which combines Transformer architectures with Cascade R-CNN for improved feature extraction and detection accuracy. In [29], Yang et al. proposed Bio-DETR, a lightweight transformer-based model designed for accurate pest and seed detection. The model integrates two self-attention-based modules, including Hybrid Scale Attention and Dynamic Bilateral Attention, to enhance feature extraction and multi-scale information fusion. In [70], Qui et al. proposed an end-to-end pest detection network that integrates residual networks with transformers by introducing a squeeze-and-excitation module for feature extraction, a multihead criss-cross attention module for efficient relational modeling, and data augmentation for class balancing.

Table 1 presents the representative deep learning methods for pest detection methods. The summary serves as a reference point to understand the methodological trends and benchmark datasets commonly adopted in recent literature.

Proposed detection method

This section presents the proposed detection method, which consists of the Attention-PestNet architecture (Section 3.1), the Spatial Downsampling Convolution (SDC) module for spatial dimension reduction (Section 3.2), the MSPA module for multi-scale feature capture (Section 3.3), and the HSDPA module for feature representation enhancement (Section 3.4).

Network architecture

Attention-PestNet utilizes a feature pyramid to make predictions at three different scales (Fig. 2). Inspired by the YOLOv8 architecture [53], our network consists of three core components: Backbone, Neck, and Head. *First*, the Backbone serves as the feature extractor, designed to capture multi-scale features from input images. The initial stage includes a SDC layer, which extracts low-level features and reduces the spatial resolution to prepare for deeper processing. The subsequent stages consist of multiple MSPA layers followed by Cross-Stage Partial with Fusion (C2f) layers. This structured design enhances feature extraction efficiency and improves the

Table 1 Summary of deep learning-based pest detection methods

Authors	Year	Method name	Architecture	Datasets
Yang et al. [16]	2025	YOLO-RP	YOLO11n	IP102
Tan et al. [22]	2025	M2DETR	Transformer	IP102, R2000
Zheng et al. [17]	2025	Paddy-YOLO	YOLOv8	IP102, R2000
Wang et al. [23]	2025	Insect-YOLO	YOLOv8m + CBAM	Insect dataset
Hakim et al. [24]	2025	Yolo-Pest	YOLOv8x + Smart trap	Three fruit fly species
Zhang et al. [25]	2025	MSFNet-CPD	VIT + CNN	CTIP102, MTIP102
Song et al. [26]	2024	RDW-YOLO	YOLOv11 + RDFBlock	IP102 modified
Zheng et al. [27]	2024	Rice pest detection	YOLOv8	IP102, R2000, IPR16
Zeng et al. [28]	2024	HCFormer	CNN + ViT	Flickr dataset
Wei et al. [13]	2024	AEC-YOLOv8n	YOLOv8 + Attention	IP102
Di et al. [29]	2024	Bio-DETR	YOLOv8 + Transformer	Pest24, HSI-Bio
Yang and Qiu [30]	2024	YOLOv5s-pest	YOLOv5 + NCBAM	IP16
Dong et al. [31]	2024	PestLite	YOLOv5 + ECA	IP102
Yang et al. [32]	2023	Imp. YOLOv7-Tiny	YOLOv7-Tiny	Tea tree pest
Peng and Wang [33]	2023	RS Transformer	CNN + Transformer	IP102, D0, Li
Zhang et al. [34]	2023	Aphid Clus. Det.	VNet, GFLV2, PAA, ATSS	Aphid images
Amrani et al. [35]	2023	CSA-RPN-PSSM	Reg-based CNN + CSA	AphColDat
Di et al. [36]	2023	TP-YOLO	YOLOv8 + CoTM + ODCM	Khapra beetle, Pest24
Tang et al. [37]	2023	Improved Pest-YOLO	YOLOv4 + ECA + CSFF	Pest24
Xiang et al. [15]	2023	Yolo-Pest	YOLOv5s + CAC3	IP102
Nanni et al. [38]	2022	CNNs	ResNet50, GoogleNet, etc.	IP102, Deng
Fan et al. [14]	2022	Agri. Pest	YOLOv5s + SE	IP102
Li and Lee [9]	2022	ResNet-101	CascadeR-CNN + Swin-T	Citrus fruits
Lyu et al. [39]	2021	Improved SSD	SSD300 + K-means	Wheat pests

representation of spatial information. *Second*, the Neck utilizes an optimized version of the Path Aggregation Network (PANet), designed to enhance the flow of information across different feature levels. Features extracted from the backbone pass through HSDPA layers. These features are then upsampled and combined with features

from previous layers using concatenation. This approach facilitates better feature aggregation and improves object detection performance at different scales. *Finally*, the Head adopts a decoupled design to performs object classification, bounding box regression, and objectness scoring in parallel, improving prediction accuracy through task-specific submodules.

Spatial downsampling convolution module

In the initial of the backbone, the SDC module extracts essential features and reduces spatial dimensions (Fig. 3). Inspired by ResNet-18 [59], the proposed module adopts a two-layer convolutional design that excludes residual connections. The first layer applies a 3×3 convolution with a stride of 2 and appropriate padding to achieve spatial downsampling while preserving critical features. Batch Normalization and the SiLU activation function are applied to enhance training stability and improve non-linear representational capacity. Subsequently, the second 3×3 convolution with a stride of 1 further refines the feature representations while maintaining the spatial resolution. Compared to the standard Conv block in YOLO [50] or the ADown [55] module, the SDC module simplifies the design by avoiding extra options (groups, dilation, autopad) and does not split channels or combine pooling branches, which reduces implementation complexity and memory overhead while maintaining essential feature extraction for high-resolution inputs.

Multi-scale spatial attention module

Inspired by Coordinate Attention [71] and SPP modules [72], we introduce the MSPA module (Fig. 4). The MSPA module integrates the multi-scale spatial attention mechanisms and multi-scale pooling to enhance feature extraction. The global average pooling is first applied separately along the horizontal and vertical dimensions. This process refines the feature representation by emphasizing the most important spatial relationships in each direction. Specifically, given the input $X \in \mathbb{R}^{C \times H \times W}$, we use two spatial extents pooling kernels of size $H \times 1$ or $1 \times W$ to encode each channel along the horizontal coordinate and the vertical coordinate, respectively. Thus, the output of the c -th channel at height h can be formulated as

$$z_c(h) = \frac{1}{W} \sum_{0 \leq i < W} x_c(h, i), \quad (1)$$

where $x_c(h, i)$ denotes the activation value at row h and column i of the c -th channel in the input feature map, and W is the width of the feature map. Similarly, the output of the c -th channel at width w can be rewritten as

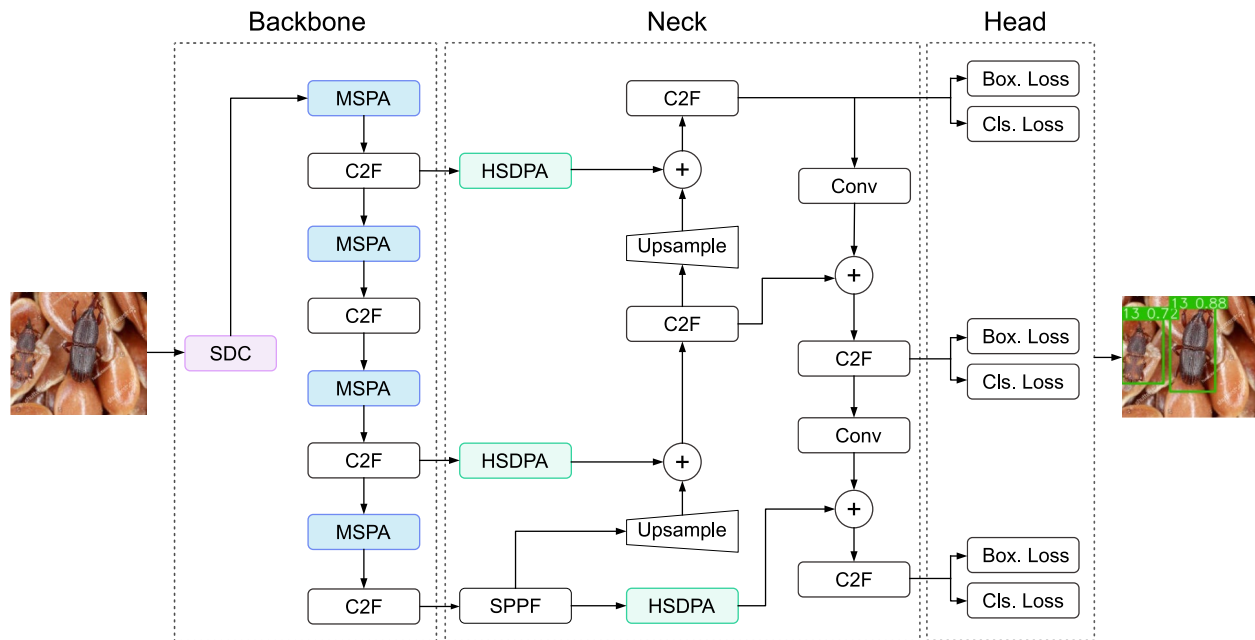


Fig. 2 Overall architecture of the proposed Attention-PestNet model for insect pest detection. The proposed modules are highlighted for clarity. Best viewed in color



Fig. 3 Detailed structure of the proposed spatial downsampling convolution module

$$z_c(w) = \frac{1}{H} \sum_{0 \leq i < H} x_c(j, w), \quad (2)$$

where $x_c(j, w)$ denotes the activation value at column j and row w of the c -th channel in the input feature map, and H is the height of the feature map.

Next, the output $Y \in \mathbb{R}^{C \times H \times W}$ is obtained by element-wise multiplication of the input X with the global average pooling features extracted independently along the horizontal and vertical dimensions. To further enrich feature representation, three max-pooling layers with

kernel sizes of 3, 7, and 13 are applied to the output Y to capture spatial features at multiple scales. Each max-pooling layer applies a specific stride to reduce spatial dimensions, while padding maintains alignment for concatenation. The output feature maps are concatenated along the channel dimension, forming a unified multi-scale representation.

Finally, a convolutional layer reduces the number of channels to a specified value, followed by batch normalization and a SiLU activation function.

The MSPA module is designed to explicitly mitigate both intra- and inter-species variation by enhancing the model's ability to capture discriminative spatial cues across multiple scales. First, the directional pooling operation encodes global dependencies along the horizontal and vertical axes, enabling the network to highlight

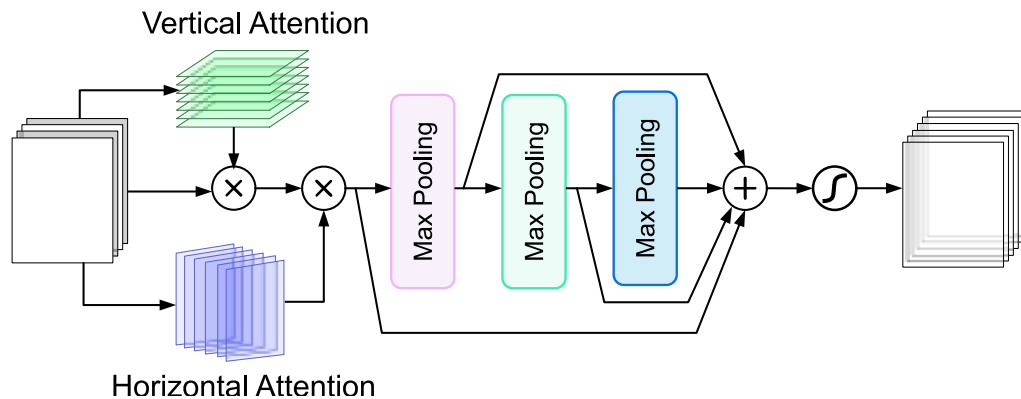


Fig. 4 Structure of the proposed multi-scale spatial attention module

stable structural patterns that remain consistent within the same species across developmental stages. Second, the multi-scale max-pooling branches with different kernel sizes allow the module to simultaneously capture fine-grained and coarse-grained spatial details. Fine-scale responses help differentiate subtle morphological cues between visually similar species, while larger receptive fields capture holistic contours and global body structures. MSPA module concatenates features from complementary scales and refines them through convolutional filtering, thereby effectively enhancing inter-species separability even when species exhibit high visual similarity. In our overall architecture, MSPA acts as the earliest stage of spatial refinement, providing variation-invariant spatial cues that stabilize the downstream attention modules. As a result, the model becomes more robust to the substantial morphological changes observed across pest developmental stages and the high visual similarity among species.

Hierarchical scale dot-product attention module

The HSDPA module is designed to explicitly address both intra- and inter-species variation by modeling global contextual dependencies at multiple representational levels. In insect pest imagery, individuals of the same species often exhibit substantial intra-species variation due to developmental stage, deformation, or changes in viewpoint. Conversely, many species share highly similar local textures or colors, making inter-species discrimination difficult. HSDPA mitigates these challenges by learning long-range, direction-aware correlations that highlight stable morphological structures while suppressing ambiguous, locally similar patterns.

Tri-Directional Attention: At the core of HSDPA is the TDA block, which factorizes attention computation along the height, width, and channel dimensions to capture structured dependencies efficiently. Unlike conventional self-attention [73], which scales quadratically with the spatial resolution, TDA decomposes the interaction space and reduces complexity from

$$\mathcal{O}(H^2W^2C) \text{ to } \mathcal{O}(HWC^2),$$

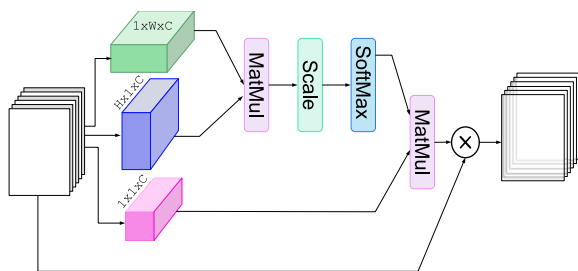


Fig. 5 Structure of the proposed tri-directional attention module

while maintaining the ability to model non-local relationships.

Inspired by the Scaled Dot-Product attention mechanism, the TDA module extends this concept to both spatial and channel dimensions. It integrates height-wise, width-wise, and channel-wise attentions into a unified structure. Specifically, it computes dot-product interactions across spatial locations, applies Softmax normalization, and incorporates channel attention as value vectors. The resulting output reweights the input features, enhancing salient patterns across both spatial and channel dimensions (Fig. 5).

In the proposed design, the input feature map $X \in \mathbb{R}^{B \times C \times H \times W}$ serves as the input to three lightweight convolutional branches corresponding to the height, width, and channel directions. The height-wise branch generates the query feature map $Q_h \in \mathbb{R}^{B \times C \times H \times 1}$, the width-wise branch produces the key feature map $K_w \in \mathbb{R}^{B \times C \times 1 \times W}$, and the channel-attention branch yields the value feature map $V_c \in \mathbb{R}^{B \times C \times 1 \times 1}$. After reshaping and flattening along the corresponding dimensions, these become:

$$\begin{aligned} Q &= \text{reshape}(Q_h) \in \mathbb{R}^{B \times C \times H}, \\ K &= \text{reshape}(K_w) \in \mathbb{R}^{B \times C \times W}, \\ V &= \text{reshape}(V_c) \in \mathbb{R}^{B \times C \times 1}. \end{aligned} \quad (3)$$

The attention score matrix is computed as

$$S = Q K^T, \quad A = \text{Softmax} \left(\frac{S}{\sqrt{d_k}} \right), \quad (4)$$

where $Q, K \in \mathbb{R}^{B \times C \times (H \times W)}$, and $d_k = H \times W$ is the scaling factor.

The value $V \in \mathbb{R}^{B \times C \times 1}$ is derived by flattening the channel attention output. The final attention output is given by

$$O = A V, \quad Y = O \odot X, \quad (5)$$

where $\odot$ denotes element-wise multiplication and $X \in \mathbb{R}^{B \times C \times H \times W}$ is the original input. This operation allows the model to refine feature representations using joint spatial and channel attention.

This directional decomposition allows the model to emphasize stage-invariant and pose-invariant structures (e.g., global contour continuity) that reduce intra-species variation, while long-range spatial interactions help distinguish species that appear locally similar but differ in overall morphology—thus improving inter-species separability.

Hierarchical Scale Dot-Product Attention Module: To further enhance representational richness, multiple TDA

blocks are stacked hierarchically. Each block extracts complementary contextual cues at different semantic depths, and their outputs are concatenated with the original input (Fig. 6). A 1×1 convolution followed by batch normalization refines the aggregated features into a compact, discriminative representation. This hierarchical design enables the model to integrate fine-grained patterns with long-range structure, providing robust discrimination across species with either high intra-species diversity or subtle inter-species differences.

In contrast to conventional CNN backbones that rely primarily on local receptive fields, the proposed HSDPA introduces long-range, structured dependencies that stabilize feature representation under deformation, scale changes, or occlusion. Furthermore, the reduced computational cost compared with full self-attention allows the module to operate effectively on high-resolution pest images, enabling richer contextual reasoning without incurring prohibitive memory overhead.

Experiments and analysis

This section presents the experimental evaluation of Attention-PestNet. Section 4.1 introduces the benchmark datasets used. Section 4.2 outlines the implementation details. Sections 4.3 and 4.4 report the detection performance of Attention-PestNet compared to state-of-the-art (SOTA) methods on the IP102 and R2000 datasets, respectively. Finally, Section 4.5 provides ablation studies to validate the effectiveness of the proposed architecture.

Benchmark datasets

For comprehensive comparisons, we conducted experiments on three widely used benchmark datasets: IP102, R2000, and Pest24.

IP102: This dataset comprises 75,222 RGB images spanning 102 pest categories, with 18,983 images specifically allocated for detection tasks. The dataset is further divided into a training set containing 15,178 images and a testing set with 3,798 images, both annotated with bounding boxes. IP102 includes eight super classes: five *Field Crops* (Rice, Corn, Wheat, Beet, and Alfalfa) and three *Economic Crops* (Vitis, Citrus, and Mango). Most images are close-ups of pests with complex backgrounds, diverse lighting, and high species variation.

R2000: This dataset is a selected subset of the IP102 rice pest dataset, designed to preserve comparable

characteristics and structural consistency. The images were sourced from commonly used online search engines, including Google, Flickr, and Bing. R2000 focuses exclusively on rice pests and comprises 2,046 images spanning 16 distinct categories. The dataset is divided into three subsets: 1225 images for training, 298 for validation, and 523 for testing.

Pest24: This dataset is a large-scale dataset designed for agricultural pest detection. It includes 25,378 publicly available images and 192,422 annotated instances. The dataset covers 24 distinct categories of crop pests, which are typically small in size, visually similar in shape, and densely distributed-posing significant challenges for accurate detection and classification.

Implementation details

The proposed method was implemented and evaluated using the Ultralytics framework [53]. For training optimization, we used the SGD optimizer with a learning rate of 0.01 and a momentum of 0.9. The input size was set to 640×640 pixels, and the batch size was 32. We trained the models for 200 epochs and selected the best-performing model on the validation set as the final version. To ensure statistical reliability, all experiments were repeated three times under identical settings, and the average results across runs were reported in the tables. All training and testing were performed on an NVIDIA GeForce RTX 4090 GPU.

Comparisons with other methods on IP102 dataset

Quantitative Results: We evaluate Attention-PestNet against seventeen state-of-the-art detection methods on the IP102 dataset. Table 2 shows the performance of the evaluated methods. In terms of detection performance, Attention-PestNet achieves mAP@50 of 68.67%, outperforming the strongest baseline YOLOv8-L by 1.81%. It also yields the highest mAP@50:95 of 44.17%, compared to 43.25% for YOLOv8-L and 42.76% for YOLOv9-C. To ensure that these differences are statistically meaningful, each model was trained and evaluated three independent times under identical experimental conditions. The mean and standard deviation of mAP values were computed across runs, and a Tukey's Honest Significant Difference (HSD) test was conducted at the 95% confidence level to compare Attention-PestNet with other methods. The results confirmed that the improvements achieved by Attention-PestNet were statistically significant ($p < 0.05$) for both mAP@50 and mAP@50:95 metrics. Traditional two-stage detectors, including Faster R-CNN, Dynamic R-CNN, and Cascade R-CNN, show limited performance, with mAP@50 values between 45.57% and 49.80% and mAP@50:95 below 35%. Transformer-based methods, such as Swin-T (47.93% mAP@50), Swin-B (50.23% mAP@50), and RT-DETR (62.45% mAP@50), yield

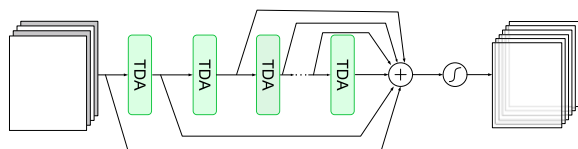


Fig. 6 Structure of the proposed hierarchical scaled dot-product attention module

Table 2 Quantitative results of different methods on the IP102 dataset

Method	mAP@50 ± SD(%)	mAP@50:95 ± SD(%)	#Params (M)	GFLOPs	Speed (FPS)
Two-stage					
Faster	49.80 ± 1.40 ^{de}	32.57 ± 1.16 ^{cd}	41.9	72.6	42.6
R-CNN [42]					
Cascade	46.40 ± 1.15 ^{bc}	33.40 ± 0.96 ^{de}	70.7	101.0	34.2
R-CNN [74]					
Dynamic	45.57 ± 1.55 ^b	27.27 ± 1.45 ^a	41.9	72.6	43.5
R-CNN [75]					
Sparse	45.53 ± 1.06 ^b	31.13 ± 1.00 ^{bc}	106.0	50.3	24.1
R-CNN [76]					
Transformer-based					
Swin-T [77]	47.93 ± 1.50 ^{cd}	30.67 ± 0.67 ^b	72.2	44.8	40.2
Swin-S [77]	51.33 ± 1.16 ^e	33.03 ± 1.10 ^{de}	98.9	66.1	24.0
Swin-B [77]	50.23 ± 1.56 ^e	32.07 ± 1.62 ^{bcd}	104.0	141.0	54.1
RT-DETR	62.45 ± 1.02 ^g	38.85 ± 1.18 ^{fg}	33.1	108.4	24.5
[78]					
Bio-DETR	55.94 ± 1.53 ^f	35.90 ± 0.29 ^e	17.9	23.7	52.1
[29]					
One-stage					
RetinaNet	49.53 ± 1.10 ^{de}	37.27 ± 1.05 ^f	38.4	216.0	14.3
[79]					
FCOS [80]	39.47 ± 1.11 ^a	26.57 ± 1.46 ^a	32.1	57.5	29.1
YOLOv8-L	66.86 ± 1.37 ^{hi}	43.25 ± 0.30 ^{ij}	43.7	165.8	39.8
[53]					
YOLOv9-C	66.07 ± 1.67 ^h	42.76 ± 1.22 ^{ij}	25.6	104.0	31.6
[54]					
YOLOv10-L	62.09 ± 1.34 ^g	39.77 ± 0.64 ^{gh}	25.9	128.1	40.7
[45]					
YOLO11-L	65.90 ± 1.45 ^h	41.39 ± 1.72 ^{hi}	25.4	87.6	38.9
[55]					
YOLO12-L	65.14 ± 1.08 ^h	41.80 ± 1.27 ⁱ	26.5	89.8	30.4
[56]					
AEC-	67.10	43.10	36.90	87.6	-
YOLOv8n					
[13]					
M2DETR	62.70	41.00	48.80	150.0	-
[17]					
Attention-PestNet	68.67 ± 1.03ⁱ	44.17 ± 0.47^j	79.6	162.7	39.2
(Ours)					

Bold values indicate the best performance in each column

SD stands for standard deviation. Different superscript letters (a–j) indicate statistically significant differences between methods according to Tukey's HSD test at the 95% confidence level. The results of YOLO-AEC and M2DETR are cited from the original papers; no statistical significance is reported for these methods in the current comparison

moderate accuracy but remain below Attention-PestNet across both metrics.

In terms of computational efficiency, Attention-PestNet achieves a well-balanced trade-off between complexity and detection performance. It requires 79.6M parameters and 162.7 GFLOPs, which is more than

lightweight models such as YOLOv8-L (43.7M parameters, 165.8 GFLOPs) and YOLOv9-C (25.6M parameters, 104.0 GFLOPs), yet it delivers superior overall performance. In addition, Attention-PestNet can operate at a speed of 39.20 FPS, which is approximately $2.7\times$ faster than RetinaNet and $1.3\times$ faster than FCOS. While Swin-T (40.2 FPS) and Bio-DETR (52.1 FPS) are slightly faster, they achieve this at the cost of significantly lower detection accuracy.

Qualitative Results: Figure 7 presents qualitative detection results on the IP102 dataset, highlighting the model's ability to handle intra- and inter-species similarity. Two groups of visually similar species are considered: Group 1 illustrates the morphological variations across developmental stages of *Corn Borer* (ID 22), while Group 2 comprises *Yellow Rice Borer* (ID 01), *Black Cutworm* (ID 17) and *Large Cutworm* (ID 20).

Within Group 1, the two-stage models fail to capture intra-species morphological differences of *Corn Borer* (ID 22), as evidenced by the unrecognized second individual in column 2. For the single-stage approach, all models except YOLOv9C misclassified the second individual. The remaining models demonstrated consistent confidence in recognizing, even when morphological variations were present compared to the reference image in column 1. This highlights their robustness in handling intra-species differences across developmental stages.

For Group 2, the proposed model exhibits enhanced inter-species separability among *Yellow Rice Borer* (ID 01), *Black Cutworm* (ID 17), and *Large Cutworm* (ID 20). In particular, our model together with YOLOv9C effectively distinguishes *Yellow Rice Borer* from *Corn Borer*, despite their pronounced morphological similarity. Moreover, although *Black Cutworm* and *Large Cutworm* share only subtle morphological differences—reflected by a relatively low confidence score of 28%—our model successfully localizes both species, whereas RT-DETR generates multiple erroneous bounding boxes for ID 17. For the less challenging case of ID 20, both our model and RT-DETR consistently achieve confidence scores above 80%, thereby outperforming one-stage and two-stage baselines. These results highlight the robustness of our approach in handling inter-species morphological overlaps and in maintaining high confidence across varying levels of classification difficulty.

These results indicate that the proposed approach effectively mitigates misclassification caused by intra- and inter-species similarity. The integration of multi-scale spatial and hierarchical attention mechanisms enables the network to capture both fine-grained morphological cues and global contextual patterns, thereby enhancing detection accuracy across morphologically diverse pest species.

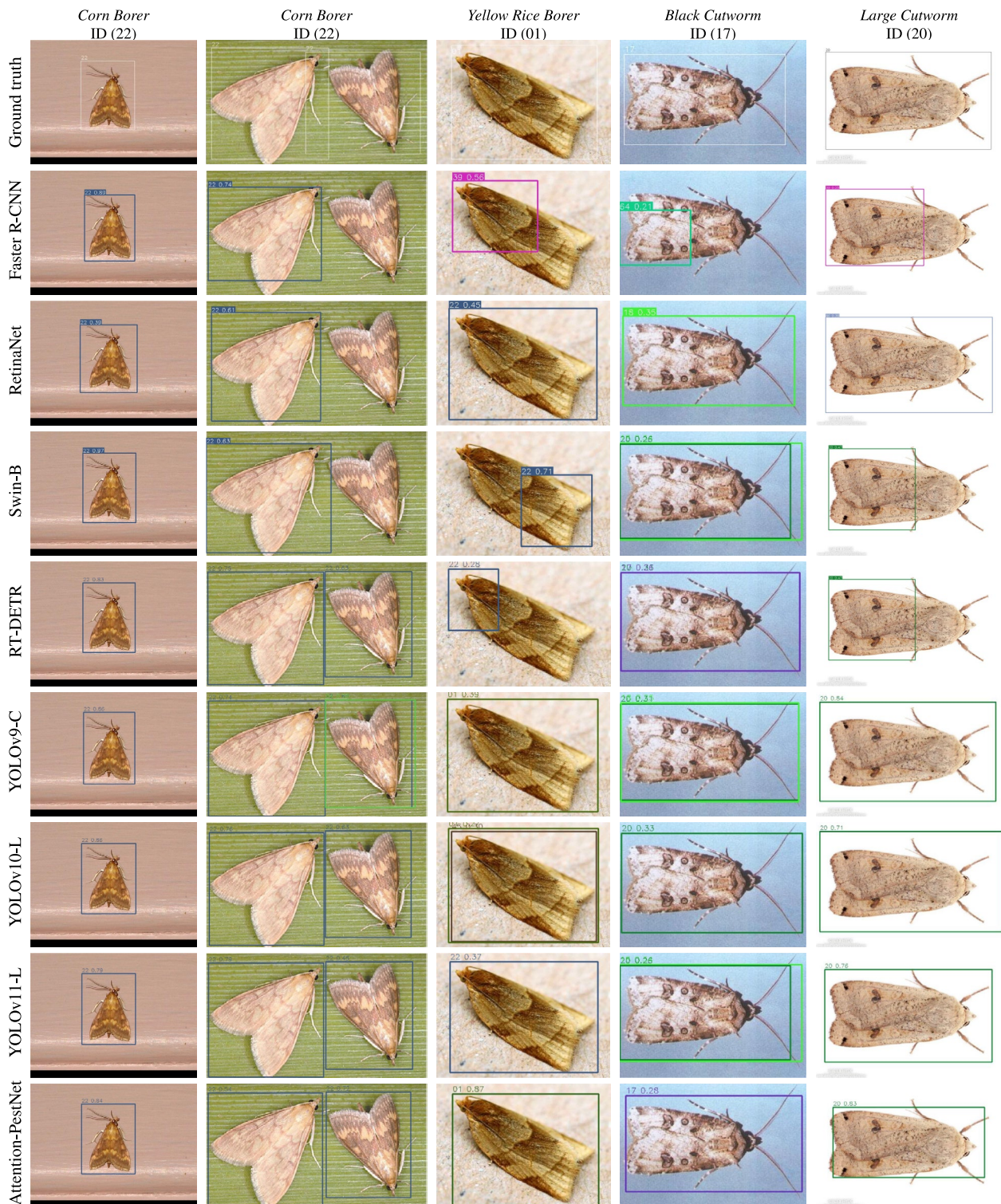


Fig. 7 Qualitative detection results on the IP102 dataset. Predicted labels show both intra-species variations and inter-species separability followed by confidence scores. The top row illustrates the class names corresponding to each detection. Best viewed in color

Comparisons with other methods on R2000 dataset

Quantitative Results: Table 3 presents the performance of different detection methods on the R2000 dataset. The results indicate that Attention-PestNet outperforms all examined models in both detection metrics. It also achieves faster inference compared to two-stage and transformer-based detectors. Our proposed method produces the best performance, with an mAP@50 of 81.54% and an mAP@50:95 of 68.00%, outperforming the second-best model YOLOv12 by 3.6% and 1.53%, respectively. Compared to the one-stage detectors such as YOLOv8-L (67.81% mAP@50) and YOLOv9-C (65.06% mAP@50), Attention-PestNet yields significant improvements. Compared to the two-stage models such as Faster R-CNN and Dynamic R-CNN, improvements of 16.11% and 20.17% mAP@50 are achieved. Furthermore, in the context of transformer-based detectors, the model demonstrates clear advantages, exceeding the performance of strong baselines such as Swin-B by 6.94% mAP@50 and 18.9% mAP@50:95, indicating its robustness in complex detection tasks.

Qualitative Results: Figure 8 illustrates the recognition performance of different models on *Mythimna pallens* (ID 01) across multiple developmental stages exhibiting distinct morphological characteristics. The

results demonstrate that both the proposed model and RetinaNet outperform other baselines in correctly classifying instances at varying stages, with our model achieving consistently higher confidence scores. Among the remaining methods, Faster R-CNN, SwinB, and YOLOv9C correctly predict 2 out of 4 stages, while RT-DETR, YOLOv10L, and YOLOv11L correctly predict only 1 out of 4, highlighting the challenges posed by intra-species morphological variation.

Notably, our model maintains robust detection across both subtle and pronounced morphological changes, suggesting that the multi-scale spatial and hierarchical attention mechanisms effectively capture stage-invariant features. This capability allows the network to emphasize stable structural patterns, such as body contour, wing shape, and relative proportions, which remain consistent despite developmental differences. Consequently, the proposed approach mitigates misclassification that typically arises when the same species presents diverse appearances, demonstrating its effectiveness in handling intra-species variability in practical pest detection scenarios.

Ablation studies

In this section, we ablate salient design elements in the proposed method. Given that the R2000 dataset is derived from IP102 and exhibits similar structural and visual features, the ablation studies were conducted only on IP102.

Number of Hierarchical Attention Levels: Table 4 analyzes how varying the number of hierarchical attention levels (denoted by k) affects the detection performance and computational cost. As the number of levels increases from 2 to 4, the model achieves a consistent improvement in performance, with mAP@50 increasing from 68.5% to 69.7% and mAP@50:95 from 43.3% to 44.9%. However, beyond $k = 4$, the performance gains plateau and slightly decline, while both the number of parameters and GFLOPs continue to increase. At $k = 6$, for instance, the model incurs higher computational demands (85.1M parameters and 190.26 GFLOPs) but demonstrates a drop in mAP@50 to 68.7%. These results suggest that increasing hierarchical depth beyond a certain point yields diminishing returns. Therefore, $k = 4$ emerges as the most effective trade-off between detection accuracy and computational efficiency.

Model Size Analysis: Based on the results in Table 4, we introduce four Attention-PestNet variants using four-level hierarchical attention, which offer an optimal trade-off between detection accuracy and computational efficiency. Table 5 summarizes the architectural configurations of four Attention-PestNet variants used in the experiments. Each model is characterized by three key parameters: depth multiplier, width multiplier, and

Table 3 Quantitative results of different methods on the R2000 dataset

Method	mAP@50 ± SD(%)	mAP@50:95 ± SD(%)
Two-stage		
Faster R-CNN [42]	65.43 ± 1.48 ^{fg}	43.03 ± 1.86 ^f
Cascade R-CNN [74]	67.73 ± 1.88 ^g	45.83 ± 0.76 ^g
Dynamic R-CNN [75]	61.37 ± 1.00 ^e	36.50 ± 0.96 ^e
Sparse R-CNN [76]	62.03 ± 1.50 ^e	48.07 ± 1.62 ^{hi}
Transformer-based		
Swin-T [77]	58.70 ± 1.78 ^d	27.35 ± 0.96 ^b
Swin-S [77]	72.37 ± 1.10 ^{hi}	50.33 ± 1.66 ^{jk}
Swin-B [77]	74.60 ± 1.14 ⁱ	49.10 ± 1.30 ^{ij}
RT-DETR [78]	71.43 ± 1.16 ^h	63.57 ± 0.67 ^m
Bio-DETR [29]	46.35 ± 1.77 ^b	33.33 ± 0.93 ^d
One-stage		
RetinaNet [79]	52.13 ± 1.46 ^c	29.63 ± 1.05 ^c
FCOS [80]	38.07 ± 1.16 ^a	24.44 ± 0.96 ^a
YOLOv8-L [53]	67.81 ± 1.76 ^g	57.53 ± 1.12 ⁱ
YOLOv9-C [54]	65.06 ± 1.44 ^f	51.49 ± 0.89 ^k
YOLOv10-L [45]	58.50 ± 1.07 ^d	45.64 ± 0.84 ^g
YOLO11-L [55]	58.09 ± 1.84 ^d	47.17 ± 0.99 ^{gh}
YOLO12-L [56]	77.94 ± 1.29 ^j	66.47 ± 1.07 ⁿ
M2DETR [27]	70.20	51.50
Attention-PestNet (Ours)	81.54 ± 1.03^k	68.00 ± 1.33ⁿ

Bold values indicate the best performance in each column

SD stands for standard deviation. Different superscript letters (a–n) indicate statistically significant differences between methods according to Tukey's HSD test at the 95% confidence level. The results of M2DETR are cited from the original papers; no statistical significance is reported for this method in the current comparison



Fig. 8 Qualitative detection results on the R2000 dataset. Predicted labels correspond exclusively to intra-species *Mythimna pallens*, with confidence scores shown. Best viewed in color

Table 4 Impacts of the number of Hierarchical attention levels on detection performance

No. levels	mAP@50 (%)	mAP@50:95 (%)	#Params (M)	GFLOPs
2	68.5	43.3	74.133	134.86
3	69.0	44.0	76.886	148.66
4	69.7	44.9	79.638	162.76
5	69.3	44.1	82.391	176.46
6	68.7	44.2	85.143	190.26

Bold values indicate the best performance in each column

Table 5 Architectural configuration of the four Attention-PestNet variants

Model	Depth multiplier	Width multiplier	Max channels
Attention-PestNetN	0.25	0.50	512
Attention-PestNetS	0.50	0.50	1024
Attention-PestNetM	0.33	1.00	512
Attention-PestNetL	0.33	1.00	1024

Table 6 Detection performance of the Attention-PestNet variants on the IP102 dataset

Model	mAP@50 (%)	mAP@50:95 (%)	#Params (M)	GFLOPs
Attention-PestNetN	68.3	43.4	62.85	127.9
Attention-PestNetS	68.4	43.9	77.51	151.1
Attention-PestNetM	69.4	44.2	79.63	158.2
Attention-PestNetL	69.7	44.9	79.63	162.7

Bold values indicate the best performance in each column

maximum channel setting. The depth multiplier scales the number of layers in the network, while the width multiplier adjusts the number of channels per layer. The maximum channel setting defines an upper bound on the number of feature channels, serving as a cap to avoid excessive model width when the width multiplier is large. Specifically, Attention-PestNetN and Attention-PestNetS share the same width but differ in depth, while Attention-PestNetM and Attention-PestNetL have greater depth and progressively wider architectures.

Table 6 shows the accuracy and computational cost of four Attention-PestNet variants. The results indicate that increasing the model size leads to higher detection performance. Attention-PestNetL achieves the best accuracy, with an mAP@50 of 69.7% and an mAP@50:95 of 44.9%. However, this improvement incurs with a significantly higher in complexity, requiring 79.63M parameters and 162.7 GFLOPs. By contrast, Attention-PestNetN achieves 68.3% mAP@50 only 62.85M parameters and 127.9 GFLOPs, making it the most efficient configuration.

Pooling Window Size Configurations: In this section, we investigate the impact of pooling window sizes in the MSPA module of Attention-PestNetL. This module employs three max pooling operations with different kernel sizes to capture spatial features across multiple scales. For instance, the combination [3, 5, 7] corresponds

Table 7 Detection performance across pooling window sizes configurations

Small	Medium	Large	mAP@50 (%)	mAP@50:95 (%)
3	5	7	67.9	43.7
3	5	9	67.5	42.8
3	5	11	67.6	43.0
3	5	13	69.0	44.5
3	7	9	68.2	44.1
3	7	11	67.5	43.7
3	7	13	69.7	44.9
3	9	11	68.0	43.1
3	9	13	67.4	42.5
3	11	13	68.1	44.1
5	7	9	68.7	43.7
5	7	11	68.3	43.9
5	7	13	68.6	43.6
5	9	11	67.7	43.0
5	9	13	69.4	44.2
7	9	11	68.1	43.2
7	9	13	67.0	42.6
9	11	13	68.2	43.9

Bold values indicate the best performance in each column

to small, medium, and large kernel sizes, respectively. Table 7 summarizes the detection performance of different pooling window size combinations used in the MSPA module, evaluated by mAP@50 and mAP@50:95. Among all configurations, [3, 7, 13] yields the highest performance, with an mAP@50 of 69.7% and an mAP@50:95 of 44.9%. By contrast, the configuration with only large kernels [9, 11, 13] reduces performance, achieving only 68.2% mAP@50 and 43.9% mAP@50:95.

Attention Map Analysis: Figure 9 illustrates the effect of feature enhancement on the model's ability to capture discriminative and detailed patterns. Each pixel in the attention maps represents the activation strength at a spatial location, with higher values indicating a stronger response to object presence. Compared with other baselines, our model achieves a clearer separation of visually similar species, including *Corn Borer*, *Yellow Rice Borer*, *Black Cutworm*, and *Large Cutworm*, as highlighted in Figure 7.

The combined MSPA and HSDPA modules enhance spatial discriminability, multi-scale robustness, and global contextual awareness. MSPA captures fine- and coarse-grained spatial cues across multiple scales, while HSDPA models long-range dependencies and stage-invariant morphological features. By integrating these complementary mechanisms, the network more effectively distinguishes species with high inter-species similarity and maintains stable recognition across different developmental stages. This design directly addresses the challenges of both intra- and inter-species variation, demonstrating the model's ability to robustly identify pests under diverse conditions.

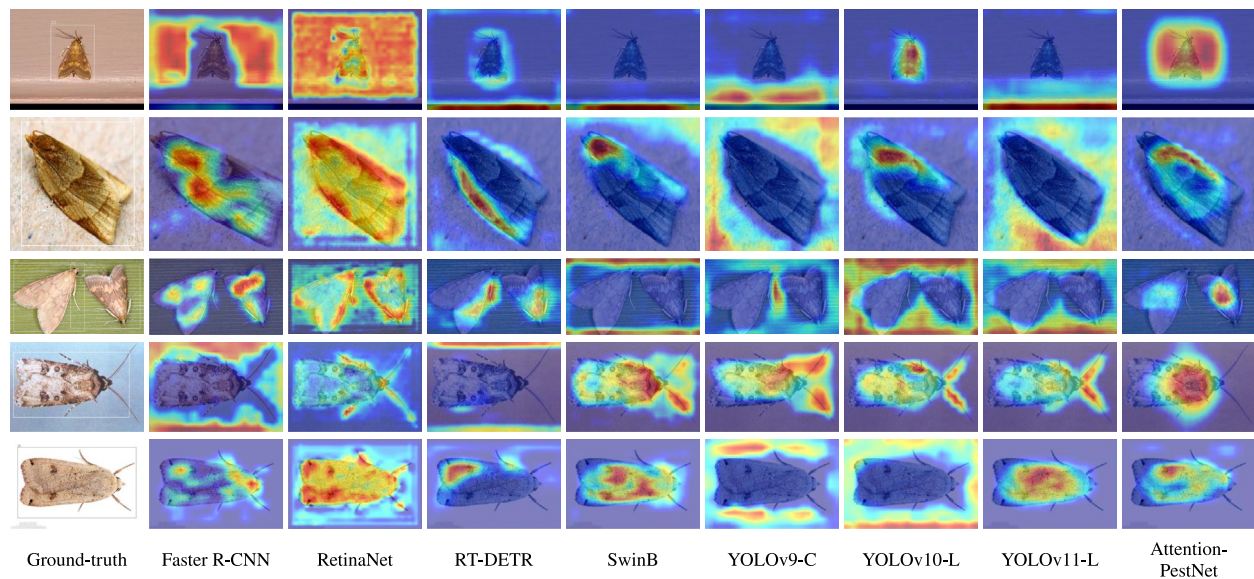


Fig. 9 Visual comparison of attention maps from different insect detection methods, highlighting both intra-species variations and inter-species separability. Best viewed in color

Table 8 Comparative analysis of spatial-channel and self-attention mechanisms on the IP102 dataset using YOLOv8L backbone

Model	mAP@50 (%)	mAP@50:95 (%)	#Params (M)	GFLOPs
Spatial-channel attention				
ECA	66.2	42.3	77.5	173.5
CBAM	66.0	42.6	53.7	127.9
CA	65.7	43.4	83.7	183.6
SE	67.0	42.9	55.7	133.0
MSPA	66.4	42.6	39.6	95.1
Self attention mechanism				
Self attention	67.0	43.3	63.7	148.3
HSDPA	67.7	44.1	75.4	117.7

Bold values indicate the best performance in each column

Comparative Analysis of Attention Modules: In this experiment, the YOLOv8L backbone is used as the base architecture, with the attention blocks replaced by channel-spatial mechanisms (ECA, CBAM, CA, SE, and the proposed MSPA module) as well as self-attention mechanisms (HSDPA and standard Self Attention). Table 8 presents the ablation results of various attention mechanisms on the IP102 dataset. Among the spatial-channel attention modules, SE Blocks achieve the highest detection accuracy, while the proposed MSPA module delivers comparable performance with the fewest parameters and lowest computational cost. In contrast, the self-attention-based mechanisms demonstrate superior feature aggregation, with the HSDPA module outperforming standard Self Attention in both accuracy and efficiency.

Comparative Analysis of Backbone Networks: Table 9 presents the detection performance of the proposed

Table 9 Comparative analysis of different YOLO backbones combined with the proposed method on the IP102 dataset

Backbone	mAP@50 (%)	mAP@50:95 (%)	#Params (M)	GFLOPs
YOLOv8	69.7	44.9	79.6	162.8
YOLOv9	67.2	44.0	28.1	142.4
YOLOv10	59.6	38.5	29.7	129.3
YOLOv11	64.0	41.0	21.7	57.3
YOLOv12	66.6	43.5	27.4	94.6

Bold values indicate the best performance in each column

method across five YOLO backbone variants. The evaluation integrates the proposed modules into YOLOv8 through YOLOv12 to examine the influence of backbone architecture. Among these variants, YOLOv8 yields the highest mAP@50 (69.7%) and mAP@50:95 (44.9%), indicating that its backbone provides the most effective feature representations. YOLOv9 and YOLOv12 achieve slightly lower scores, while YOLOv10 and YOLOv11 show lower detection performance. Based on these findings, YOLOv8 is adopted as the default backbone for all subsequent ablation studies, cross-dataset evaluations, and final performance comparisons.

Comparative Analysis of Detection Methods on the Pest24 Dataset: To evaluate the generalization of Attention-PestNet, we compared it with five representative YOLO variants on the Pest24 dataset [81], using the same training settings as in the IP102 and R2000 experiments. Table 10 shows that the proposed method achieves competitive performance compared to the existing approaches. Attention-PestNet performs slightly lower than the top-performing YOLOv8-L and TP-YOLO, but higher than YOLOv10-L, YOLOv12-L, and YOLO-RP. It achieves 66.30% mAP@50 and 42.20%

Table 10 Comparative analysis of different YOLO variants and the proposed method on the Pest24 dataset

Backbone	mAP@50 (%)	mAP@50:95 (%)
YOLOv8-L [53]	66.70	43.02
YOLOv9-M [54]	65.90	42.38
YOLOv10-L [45]	63.88	41.03
YOLOv11-L [55]	66.47	43.05
YOLOv12-L [56]	63.44	40.35
YOLO-RP [16]	58.28	35.73
TP-YOLO [36]	66.80	42.00
Attention-PestNet (Ours)	66.30	42.20

Bold values indicate the best performance in each column

mAP@50:95, demonstrating competitive accuracy among modern YOLO architectures. Although Attention-PestNet performs well on Pest24, it would be useful to further improve detection of small-scale pests and densely packed targets by investigating more refined feature extraction or attention mechanisms.

Discussion on potential improvements

The proposed Attention-PestNet has shown several advantages on multiple benchmark datasets, but our method can still have three potential improvements regarding performance for failure, trade-offs in model complexity and deployment, and the integration of multimodal data in Attention-PestNet.

Analysis of Failure Cases: Although Attention-PestNet has shown several conceptual merits, it can be further improved for detecting small-scale objects that appear densely within an image. The model tends to prioritize larger instances, potentially overlooking smaller ones (Fig. 10a–c). This issue may arise from aggressive downsampling operations in the deeper layers of the network, which substantially degrade spatial resolution. In addition, PestNet tends to merge closely positioned small objects into a single detection (Fig. 10d–f). A possible explanation for this behavior is the model's sensitivity to highly similar features, which is potentially influenced by the Scale-Dot Attention mechanism within the HSDPA module. Although this attention strategy improves generalization by emphasizing scale-consistent patterns, it may reduce the model's ability to distinguish between closely positioned small objects in crowded scenes.

To overcome these issues, future work will integrate context-aware attention mechanisms that combine local convolutional context with global self-attention [36]. This approach will dynamically adjust receptive fields based on local image structure, thus the model can capture fine details and distinguish closely spaced small objects. Additionally, dynamic kernel adaptation will improve feature extraction across varying scales, enabling more precise recognition of dense, small instances while maintaining robustness to larger patterns.

Trade-offs in Model Complexity and Deployment: Although Attention-PestNet demonstrates strong detection performance across diverse pest species, its computational efficiency could be further improved. To enhance feature representation, the model currently integrates multiple attention-based SPA and HSDPA modules. While effective, these components increase model complexity and resource usage, which may limit deployment on edge devices with constrained computing power. To address this trade-off, future work will focus on optimizing the architecture of Attention-PestNet by exploring lightweight variants. Specifically, we aim to remove redundant attention layers and adopt more efficient transformer modules to significantly reduce computational cost while preserving detection accuracy.

Multimodal Attention-PestNet: Although RGB imagery is widely used in pest detection tasks, its inherent limitations may hinder model generalizability in real-world agricultural environments. RGB sensors often fail to capture distinctive pest features under low-light conditions, occlusions, or cluttered and complex backgrounds, which are frequently encountered in the field. These limitations may reduce the effectiveness of Attention-PestNet, especially in detecting small pests, a challenge commonly observed in object detection tasks [82]. To address this issue, Attention-PestNet will be enhanced through multimodal fusion by integrating thermal and hyperspectral imaging. Based on recent advances [83], we will develop adaptive feature extractors and fusion strategies to effectively combine RGB, thermal, and hyperspectral data. In addition, meta-learning techniques [84] will also be incorporated into Attention-PestNet to enable rapid generalization to novel pest types with limited annotated samples, thereby enhancing detection performance across diverse agricultural environments. Furthermore, inspired by recent work on detecting tea leaf blight under overcast and low-light conditions using UAV-based imaging and tailored preprocessing to mitigate illumination challenges [85], Attention-PestNet will integrate robust illumination normalization and low-light enhancement modules. This will ensure that multimodal fusion remains effective even in suboptimal lighting scenarios, further improving reliability in real-world deployments.

Conclusion

In this paper, we introduce Attention-PestNet, a novel one-stage object detector specifically designed for insect pest detection. The network integrates three self-attention-based modules to enhance multi-scale feature representation and spatial awareness: (i) the MSPA module, which captures multi-scale contextual relationships to better distinguish small pests from complex backgrounds; (ii) the TDA module, which jointly models height, width, and channel-wise dependencies to refine

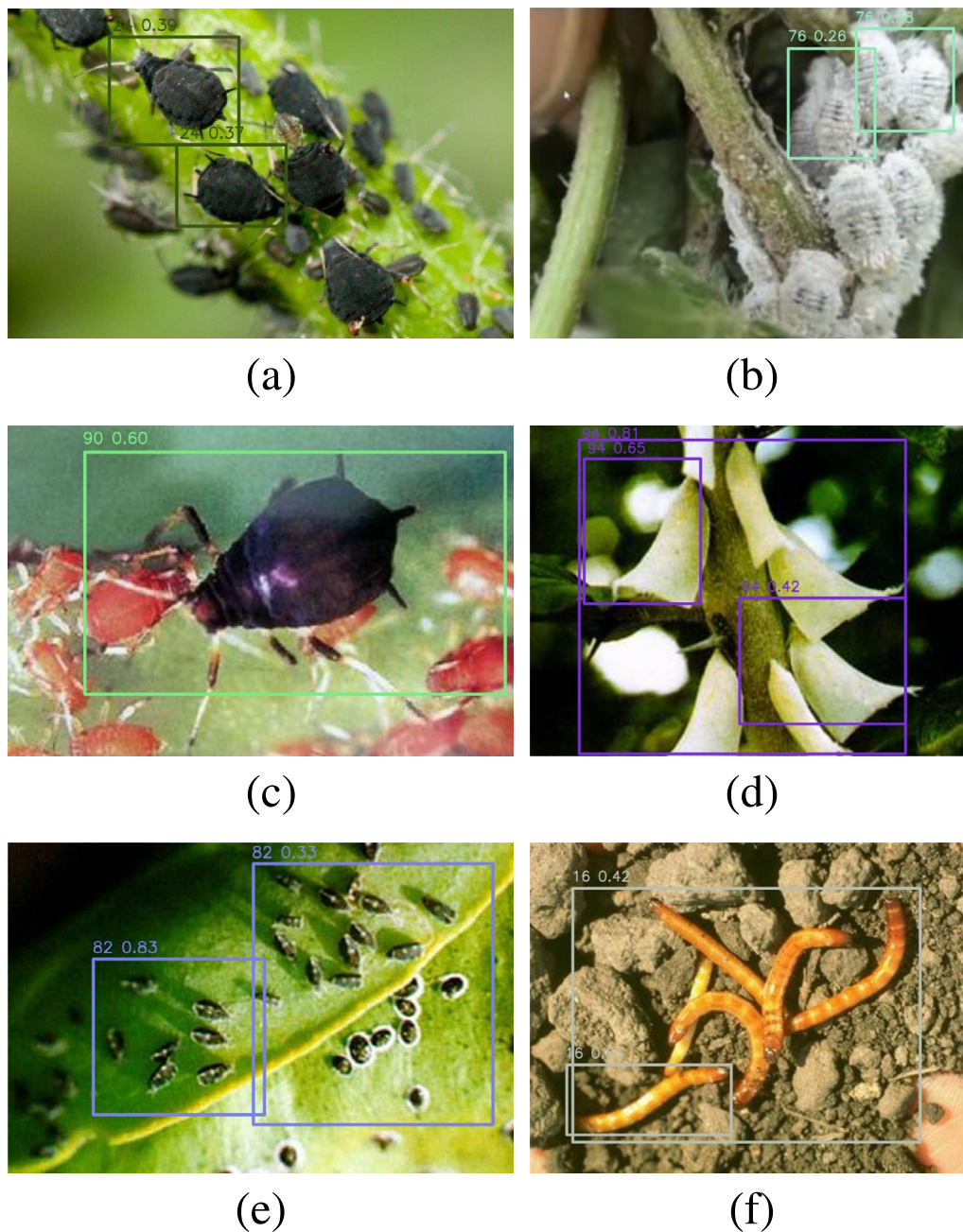


Fig. 10 Examples of the failed predictions. Predicted labels show class IDs followed by confidence scores. Best viewed in color

feature representations while reducing computational cost; and (iii) the HSDPA module, which captures long-range global dependencies to enrich semantic understanding. We evaluated Attention-PestNet on three benchmark datasets, namely IP102, R2000, and Pest24. The results show that Attention-PestNet outperforms existing detectors on IP102 and R2000, while achieving competitive performance on Pest24. Attention-PestNet achieves a favorable balance between model complexity and detection accuracy, making it practical for real-time

applications in precision agriculture on GPU-based systems.

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Author contributions

VTD: conceived and designed experiments, implemented the algorithms, wrote the original draft. HTL: contributed to design, performed analysis, and revised the manuscript. TTTP: supported investigations and data curation. HJD: acquired funding, supervised and critically revised the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets used in this study are publicly available. The IP102 dataset can be accessed at <https://doi.org/10.1109/CVPR.2019.00899>, The R2000 dataset can be accessed at <https://doi.org/10.1038/s41598-024-81587-5>. The Pest24 dataset can be accessed at <https://doi.org/10.1016/j.compag.2020.105585>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors consent to the publication of this paper.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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